MAR-PE-13 Hardware/Environmental Fault Resolution on Cisco IOS-XE PE Device (ALARM_7539)

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_7539

Description

The MAR-PE-13 node failure on the Marathahalli Core site is attributed to the occurrence of Hardware/Environmental Faults in the Cisco IOS-XE PE device. The specific failure domain of ASIC memory parity errors and line card crashes is a direct result of the intricate interplay between the device's hardware components and its operating environment. The Cisco ASR 9000 series router, being the PE (Provider Edge) device in question, relies heavily on its Line Cards (LCs) to facilitate packet forwarding and processing. The Line Cards, in turn, employ ASICs (Application-Specific Integrated Circuits) to accelerate packet processing and improve overall system performance. However, the occurrence of ASIC memory parity errors can be indicative of a hardware fault or an environmental anomaly, such as power supply redundancy loss, which can compromise the integrity of the system. Furthermore, the loss of line card redundancy can exacerbate the situation, leading to system instability and eventually a crash. To mitigate such failures, it is essential to perform a thorough diagnostic analysis and implement corrective measures to rectify the root cause of the issue.

Pre-requisite Checks

- Verify the physical integrity of the MAR-PE-13 node, including the Line Card and power supply units.

- Conduct a site survey to ensure that the environmental conditions at the Marathahalli Core site are within the recommended operating parameters for the Cisco ASR 9000 series router.

- Check the status of redundant links and ensure that they are operational and functioning as expected.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware fault or environmental anomaly	Medium
Line card crashes due to power supply redundancy loss	High
System instability due to loss of line card redundancy	Low

Diagnostic Commands & Expected Outputs

```
Command: show tech-support, Expected Output: Output from show tech-support command will include details about the device's hardware and software configuration, including Line Card and power supply information., Description: This command will provide a comprehensive view of the device's technical support information, including the Line Card's status and power supply redundancy levels.
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Command: show controller t3 0/0/0/0, Expected Output: Output from show controller t3 command will include details about the Line Card's status and ASIC memory parity errors., Description: This command will provide detailed information about the Line Card's status, including ASIC memory parity errors, which can help identify the root cause of the issue.

Command: show environment, Expected Output: Output from show environment command will include details about the device's environmental conditions, including temperature and power supply levels., Description: This command will provide a comprehensive view of the device's environmental conditions, including temperature and power supply levels, which can help identify any potential anomalies or issues.
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Resolution Steps

Step 1: Step: Replace the faulty Line Card, Command: no controllers t3 0/0/0/0, Description: This command will remove the faulty Line Card from service, allowing for its replacement with a new or refurbished unit.

Step 2: Step: Verify power supply redundancy levels, Command: show power supply, Description: This command will provide detailed information about the power supply levels, including redundancy levels, which can help ensure that the system is operating within recommended parameters.

Step 3: Step: Implement corrective measures to rectify environmental anomalies, Command: show environment, Description: This command will provide a comprehensive view of the device's environmental conditions, including temperature and power supply levels, which can help identify any potential anomalies or issues that need to be addressed.

Post-Check Verification

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Command: show tech-support, Expected Output: Output from show tech-support command will confirm that the device is operating within normal parameters and that the Line Card replacement or corrective measures have been successful., Description: This command will provide a comprehensive view of the device's technical support information, including the Line Card's status and power supply redundancy levels, which can help confirm that the resolution steps have been successful.

Command: show controller t3 0/0/0/0, Expected Output: Output from show controller t3 command will confirm that the Line Card is operating within normal parameters and that ASIC memory parity errors have been resolved., Description: This command will provide detailed information about the Line Card's status, including ASIC memory parity errors, which can help confirm that the resolution steps have been successful.
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Roll-back Procedures

- Step: Revert the Line Card replacement or corrective measures, Command: controllers t3 0/0/0/0, Description: This command will reinsert the faulty Line Card into service, allowing for the rollback of the resolution steps in the event of a failure or issue.